



COLLEGE OF ELECTRICAL AND MECHANICAL ENGINEERING NUST

MECHATRONICS DEPARTMENT

DEGREE - 45

SYNDICATE-A

MODELING AND SIMULATION

(M&S-321)

MODULE 1 REPORT

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This report presents the selection and justification for the mathematical modeling and simulation of a Lane Keeping Assist (LKA) system. The selected system has a monocular camera for line detection, a Linear Quadratic Regulator (LQR) control algorithm and a dynamic bicycle vehicle dynamics model. For the reference vehicle we have chosen the Toyota Corolla Altis X 1.6 Manual.

1. Introduction

An LKA system is critical component of modern Driver Assistance Systems (DAS). LKAs help to maintain the vehicles position within lane boundaries. The objective of this project is to develop a mathematical model and MATLAB simulation of an LKA system for autonomous vehicle steering control.

Module I will focus on selecting appropriate sensor technology, control algorithm, vehicle dynamics modeling and a reference vehicle, the technical specifications of which we can access.

2. Sensor Technology Selection

For the sensor we had a couple of options, you will find a table below that provides a good overview of the options that were considered.

Criteria	Monocular	Stereo	LiDAR	Radar
Cost	Very Low	Low-Medium	Very High	Medium
Lane Detection	Excellent	Excellent	Indirect	Poor
Depth Perception	None	Good	Excellent	Good
Weather Robustness	Poor	Poor	Moderate	Excellent
MATLAB Modeling	Very Easy	Easy	Complex	Not Suitable
Academic Literature	Abundant	Good	Moderate	Limited

Selected Sensor: The monocular camera has been selected.

Justification:

- It outputs the lateral deviation and heading error directly, which are the inputs required for the control algorithm
- It is used in MATLAB official LKA simulation examples which ensure compatibility and availability of reference material
- There is an abundance of monocular lane detection algorithms with 95-97% accuracy.

Limitation: The sensor lacks depth perception and performs poorly in adverse weather conditions. Thus, this simulation assumes ideal daylight conditions on clear, well-marked roads, which is a standard academic assumption.

3. Control Algorithm

You will find the comparative analysis of three different algorithms in the table below:

Criteria	PID	LQR	MPC
Mathematical Complexity	Low	Medium	High
Tuning Parameters	3	Matrix Q, R	Model + Constraints
MATLAB Requirements	No toolbox	Control Toolbox	MPC Toolbox
LKA Performance	Good	Better	Best
Handles Constraints	No	No	Yes
Speed Robustness	Poor	Moderate	Excellent
Academic Literature	Abundant	Good	Abundant

Selected Controller: LQR is the selected algorithm

Justification:

- Provides mathematical optimal control by minimizing the quadratic cost function

$$J = \int_0^{\infty} (x^T Q x + u^T R u) dt$$

- It is superior to PID for LKA as has been proven in literature and simpler to implement than MPC
- Works seamlessly with dynamic bicycle model's state variables

4. Vehicle Dynamics Model

Criteria	Kinematic Bicycle	Dynamic Bicycle
Speed Range	>15 m/s	5 – 25 m/s
Tire Slip Modeling	No	Yes
Inertial Effects	Negligible	Included
Parameters Required	3 (L, l_f, l_r)	7 (+ m, I_z, C_f, C_r)
MATLAB Support	Built-in block	Manual derivation
LKA Accuracy	Simplified	Realistic
Mathematical Form	Geometric	2-DOF differential equations

Selected: The dynamic bicycle model is selected.

Justification:

- For highway speeds (15-25 m/s), this model is better suited.
- Provides four natural state variables that integrate directly with LQR formulation
- Explicitly recommended over kinematics models for lateral control at highway speeds in MATLAB documentation.

5. Reference Vehicle Selection

Selected Vehicle: Toyota Corolla Altis X 1.6 manual

Justification:

- Public availability of dimensions, mass, wheelbase and other characteristics required
- Available locally, ensuring regional relevance
- Fully capable of sustained highway speed operations
- Based on well-documented vehicle dynamics in academic literature

Technical Specifications:

Below you find the technical specifications of the selected reference vehicle.

Parameter	Value
Model Year	2026
Engine	1.6L Dual VVT-i (1598 cc)
Power	120 HP @ 6000 RPM
Torque	154 Nm @ 5200 RPM
Kerb Weight (m)	1275 kg
Wheelbase (L)	2700 mm
Length × Width × Height	4620 × 1775 × 1475 mm
Ground Clearance	175 mm
Turning Radius	5.4 m
Front Suspension	MacPherson Strut with Coil Springs
Rear Suspension	Torsion Beam with Coil Springs
Tire Size	195/65 R15

6. Conclusion

This report has established and justified the configuration for an LKA system's mathematical modeling and MATLAB simulation. For a quick summary:

- **Sensor:** Monocular Camera for direct line marking detection
- **Controller:** LQR for optimal state feedback control
- **Dynamics Model:** Dynamic bicycle model for highway speed accuracy
- **Reference Vehicle:** 2026 Toyota Corolla Altis X 1.6 Manual

All assumptions have been mentioned in respective sections and justification provided. The configuration is grounded in academic literature and supported by MATLAB toolboxes. This completes our Module I report.